

Kamran Amini

Senior Systems & Solutions Architect
Independent Consultant

- ✓ CTO Advisory & Strategic Decision Making
- ✓ Systems & Solutions Architecture
- ✓ Blockchain & AI Adoption
- ✓ Data & Network Security

Author of Books & Articles on Software Engineering

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SUMMARY

With over 20 years of professional experience across software development and systems architecture, I have built and led the design of cloud-based distributed systems that meet exacting requirements for security, availability, robustness, and fault tolerance. Grounded in a computer science background, I have combined hands-on engineering with architectural leadership, remaining directly involved in coding critical components to ensure my architectural decisions are informed by implementation reality. My core expertise lies in systems-level programming languages, including C, C++, Golang, and Rust, complemented by extensive DevOps experience spanning Bash, Docker, Kubernetes, Helm, and AWS/GCP infrastructure. I have also worked extensively in start-up environments, building and scaling backend services and APIs under fast-moving product timelines, and have advised blockchain start-ups on the design and implementation of complex on-chain financial logic using Solidity. Beyond hands-on delivery, I consult companies on systems and solutions architecture, CTO-level technology strategy, blockchain adoption, and the integration of AI into engineering and business processes.

CAREER

★ ActiveVideo BV *(April 2019 - February 2023)*

↳ Commscope Inc. *(March 2023 - March 2025)*

↳ ARRIS BV *(April 2025 - December 2025)*

↳ Vistance Networks BV *(January 2026 - Present)*

Senior Software Engineer

The Netherlands, Hilversum

- CloudTV (Video Stitching Engine): Cloud HTTP renderer/streamer broadcasting menus, UI, and audio/video to set-top boxes, built on Java, C++, WebKit, and Perl.
- AppCloud: On-demand containerized Android streaming platform, with its microservices backend deployed on Kubernetes (AWS EKS).
- Multiview Engine: Cloud HTTP renderer/streamer broadcasting multiple channels simultaneously to set-top boxes and TVs.
- Member of Tech Council: Prioritized tech debt against product objectives via a dedicated backlog and Kanban flow.
- Founding member of Security Guild: Devised and enforced security standards across teams, and supported product certification.
- C++17 (g++, clang) - Golang - Java, and Kotlin - Python - AWS - Kubernetes (EKS) - S3 - EFS - EBS - Ceph - Envoy

★ TomTom BV

Senior Software Engineer *(Sept. 2017 - March. 2019)*

Netherlands, Amsterdam

- Expert Software Consultant from Modis B.V.
- NavKit Free Text Search Engine, Navkit Worker REST Services, POI Info Services
- C++11 (g++, clang) - Java SE - JUnit - Spring MVC - JAXB - Spring REST - gdb - Valgrind - Clang ASAN & TSAN - STL & Boost - Google Test - Protobuf - Thrift - Python - Jenkins - QuickBuild - Docker - JIRA - Fisheye - Confluence - Bitbucket Stash - AWS/Azure based cloud

★ Jeppesen Systems (a Boeing subsidiary)

Senior Software Engineer *(Feb. 2017 - Aug. 2017)*

Sweden, Gothenburg

- Manpower Application
- C++11 (g++) - Python - C#.NET - gdb - Valgrind (callgrind, massif) - PDB - STL & Boost - Google Test - XMLRPC - Jenkins - JIRA - Fisheye - Confluence

★ Adro Co.

Senior Software Architect, Principal C++ Engineer (*Jun. 2016 - Jan. 2017*)

Iran, Tehran

- Realtime Bidding (RTB) Engine

- C++ (g++) - Python - gdb - Valgrind - STL & Boost - libuv - Google Test - Protobuf - Jenkins - Travis CI - Docker - JIRA - Fisheye

★ Naeem Co.

Chief Technical Officer, Chief Software Architect (*Apr. 2015 - Oct. 2016*)

Iran, Tehran

- Projects: ePassport Issuance System - Face Recognition System, Biometric Identification Sys.

- Crypto.: OpenSSL - PKCS (1, 7, 10, 11, 12) - ICAO MRTD standards - HSM - Key Mgmt. - PKCS11 Tokens

- Stack: C (gcc) - C++ (g++) - Java SE - Java EE - Spring MVC - Wildfly - Tomcat - Python - gdb - Valgrind - STL & Boost - wxWidgets - Qt - Google Test - Jenkins - Docker - vSphere - Proxmox - Gluster - Cassandra - Redmine - GitLab

★ Sharif CERT Center

Software Engineer (*Dec. 2011 - Jun. 2013*)

Iran, Tehran

- Botnet Detection System - Sinkhole - DDNS as a FreeBSD module

- C++ (g++) - Java SE - Perl - gdb - JConsole - Valgrind - STL & Boost - Swing - Redmine

TECHNICAL SKILLS

- **Programming Languages**: C++ (14, 17, 20) - C - Golang - Rust - JS (ES5, ES6) - Java (11+) - Python (2, 3) - Perl
- **Blockchain Development**: Solidity (EVM chains) - Solana - web3 (ethers.js, web3.js) - Hardhat
- **Architecture Concepts**: ACID - CAP - In-memory and Dist. caches - Distributed key-value stores - Replication - Fault tolerance - Leader election - Dist. locks - Dist. file systems - Streaming - Micro-batching - CQRS - Event Sourcing - Message Queues - Green threads - Event loops and async programming - Load balancers
- **Data Storage**: PostgreSQL - MySQL - MongoDB - Redis - Dist. ledgers (blockchains)
- **Kubernetes**: kubectl - k9s - Helm - Kustomize - Configuring resources such as ingress, pod, deployment, replica set, secrets, pod security policies, pod security admission (PSA) as of 1.25 - Migration from earlier versions to 1.23 and 1.25
- **Amazon AWS & Google Cloud (GCP)**: awscli - eksctl - gcloud - Configuring resources such as EC2 instances, EBS volumes, EFS, S3, ALBs, EKS, IAM, GCE, GKE, Cloud Storage, VPC, etc
- **Serialization & RPC**: Protobuf - gRPC - ZeroMQ - Thrift - JSON over REST - SOAP - Java RMI - XMLRPC
- **Standards**: SUS and POSIX - PKCS standards - JavaEE (JPA, JSM, EJB, Java Servlets, JAX-WS, JAX-RS, etc) - RFCs (ASN.1, IP, TCP, UDP, DNS, HTTP, CMS, etc), ERCs (20, 721, 1155, 165, 2981, 2535, ...)
- **Cryptography**: OpenSSL - Symmetric algos (DES, 3DES, AES) - Asymmetric algos (RSA, ECC, DH) - SSL - PKI (CA, Digital Signature, PKCS certs, HSMs and Crypto. Tokens) - Wrap Keys - Key Derivation Algorithms - Hash algos - MAC algos - Error Detection (Crypto. Hashes), studying for CISSP certificate
- **Computer Networks**: OSI & Internet Protocol Suite, Packet Switching, Error Detection & Correction (Checksum, CRC, ARQ, FEC), L3 protocols (ICMP, IP) - L4 protocols (UDP, TCP) - Socket Programming - L3 programming (libpcap, JNetPcap) - Wireshark analysis - Penetration Testing - tcpdump
- **Kernel Development**: Linux Modules - Linux System Calls
- **Machine Learning**: CUDA Dev. - Tensorflow (C++ & Python) - OpenCV (C, C++, Python) - Supervised & Unsupervised Learning - Reinforced Learning - Deep Learning - Locality Sensitive Hashes - Iris Code - Gabor filter - Feature Selection & Extraction

Technical authorship & open source projects

- **Extreme C**: My first book was published in October 2019 by Packt Publication.
- **Linux Two-Step Authentication with Rust**: A liveProject published by Manning Publications that walks through the students and makes them able to use Rust to develop a two-step verification system for authentication services.
- **Encryptizer Simurgh Software (ESS) Backend Project**: Rust RESTful backend written for the Manning liveProject
- **Encryptizer Simurgh Software (ESS) Clients Project**: Rust REST client written for the Manning liveProject

- **Domain Catcher Engine:** Written in Rust meant to send EPP request to catch the dropped domains
- **Outlook Email Message (.msg) parser:** Written in Rust

FOREIGN LANGUAGES

Persian (**Native**), English (**Fluent**), French (**A1**), Dutch (**A2**)

EDUCATION

- (2017 - Present) **Bachelor of Science in Physics (Astronomy and Planetary Science)** at The Open University
- (2004 - 2008) **Bachelor of Software Engineering** at University of Isfahan (Isfahan, Iran)

PERSONAL INTERESTS

- **Physics and Astronomy:** Cosmology, Astronomy, Quantum Physics, Quantum Mechanics, Particle Physics, Special and General Relativity, Theory of Everything, Photonics.
- **Mathematics:** Linear and Abstract Algebra, Probability Theory, Information Theory
- **Cryptography:** Information Security, Network Security, Security Algorithms, Quantum Information Theory, Quantum Key Distribution, Quantum Cryptography
- **Philosophy:** Western Philosophy, Sociology, Economics, Politics and Humanities
- **Telecommunications:** Modulation, Multiplexing, Codes, Telecommunication circuits, Photonics and optical fibers, IEEE 802.3 Protocols, Information Theory, Coding Theory, Computer Networks, etc